GeoModels Tutorial: analysis of spatial precipitation anomalies

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Introduction

In this tutorial we show how to analyze total precipitation anomalies registered at 7,352 location sites in the USA, from 1895 to 1997. A detailed description of the data can be found in Kaufman et al. (2008). The yearly totals were standardized using station-specific long-run means and standard deviations. The dataset is large, and this tutorial uses `GeoModels` (Bevilacqua et al., 2026) to estimate and predict with Gaussian and skew-Gaussian random fields. We first load the required R libraries.

```
rm(list=ls())
require(GeoModels)
require(fields)
require(mapproj)
require(maps)
require(sp)
require(geoR)
```

1 Preliminary data analysis

Precipitation anomaly data are available in the `GeoModels` package.

```
data("anomalies",package="GeoModels")
head(anomalies)
          lon    lat           z
 [1,]  -85.25  31.57   -0.4586873
 [2,]  -87.42  32.23   -0.9253283
 [3,]  -85.87  32.98   -0.4370817
loc_lonlat=cbind(anomalies[,1],anomalies[,2])
z=as.numeric(anomalies[,3])
```

The coordinates are expressed in longitude and latitude (decimal degrees). A map of the anomalies (Figure 1) is obtained from:

```
quilt.plot(loc_lonlat,z,xlab="long",ylab="lat")
map("usa",add=TRUE)
```

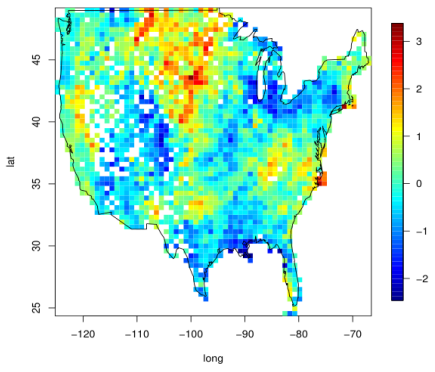


Figure 1: Coloured map of anomalies data.

We first project the spherical coordinates onto a two-dimensional Euclidean plane using the sinusoidal projection. GeoModels can also handle longitude–latitude coordinates with geodesic or chordal distances, but this example uses projected coordinates (in kilometres).

```
P.sinusoidal <- mapproject(loc_lonlat[,1],loc_lonlat[,2],
                           projection="sinusoidal")
loc <- cbind(P.sinusoidal$x,P.sinusoidal$y)*6371
maxdist <- max(dist(loc[chull(loc),,drop=FALSE]))
```

Here 6371 is the assumed radius of the Earth in kilometres. Calculating the maximum pairwise distance from the convex-hull vertices gives the same diameter as the full point cloud, without constructing its full distance matrix. The marginal histogram suggests that a Gaussian assumption may be reasonable, although there is evidence of slight asymmetry. The nearest-neighbour scatterplots provide information about bivariate dependence (Figure 2).

```
summary(z)
hist(z,main="Anomalies histogram")
GeoScatterplot(data=z,coordx=loc,neighb=c(1,2,3,4))
```

The empirical semivariogram (Figure 3) suggests a non-negligible nugget effect:

```
evariog=GeoVariogram(data=z,coordx=loc,maxdist=maxdist/4)
plot(evariog,ylim=c(0,1),pch=20,
     xlab="Km",ylab="Semi-variogram")
```

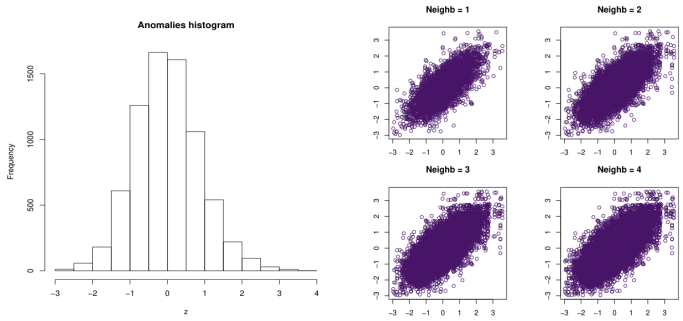


Figure 2: Histogram of anomalies and nearest-neighbour scatterplots.

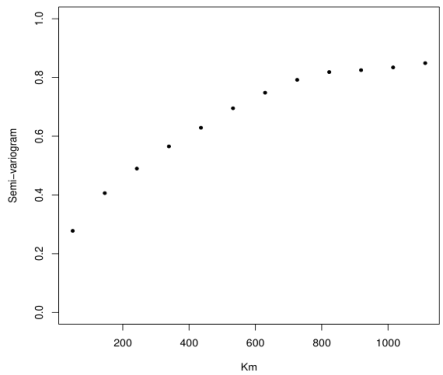


Figure 3: Empirical semivariogram of anomalies.

The preliminary analysis motivates comparing Gaussian and skew-Gaussian random fields with a nugget effect.

2 Gaussian and skew-Gaussian random fields

Consider a standard stationary Gaussian random field $G = \{G(\boldsymbol{s}), \boldsymbol{s} \in \mathcal{S} \subset \mathbb{R}^2\}$ with zero mean, unit variance, and isotropic exponential correlation. The next section summarizes the two models.

The correlation model, with nugget parameter $\tau^2 \in [0, 1)$, is

$$\rho(\boldsymbol{h}) = \begin{cases} 1, & \|\boldsymbol{h}\| = 0, \\ (1-\tau^2)\exp(-\|\boldsymbol{h}\|/\alpha), & \|\boldsymbol{h}\| > 0, \end{cases} \qquad \alpha > 0. \tag{1}$$

The Gaussian model is the location–scale transformation

$$Y(\boldsymbol{s}) = \mu(\boldsymbol{s}) + \sigma G(\boldsymbol{s}), \qquad \mathrm{E}\, Y(\boldsymbol{s}) = \mu(\boldsymbol{s}), \quad \mathrm{Var}\, Y(\boldsymbol{s}) = \sigma^2. \tag{2}$$

The skew-Gaussian field (Zhang and El-Shaarawi, 2010) can be represented as

$$U_\eta(\boldsymbol{s}) = \mu(\boldsymbol{s}) + \eta|G_1(\boldsymbol{s})| + \sigma G_2(\boldsymbol{s}), \tag{3}$$

where G_1 and G_2 are independent copies of a standard Gaussian random field. We take G_1 without a nugget and let the nugget in G_2 be as in (1). Its marginal mean and variance are

$$\mathrm{E}\, U_\eta(\boldsymbol{s}) = \mu(\boldsymbol{s}) + \eta\sqrt{2/\pi}, \qquad \mathrm{Var}\, U_\eta(\boldsymbol{s}) = \sigma^2 + \eta^2(1 - 2/\pi).$$

In particular $\eta = 0$ gives the Gaussian model. Writing $\rho_1(\boldsymbol{h})$ for (1) with zero nugget, the skew-Gaussian correlation function is

$$\rho_{U_\eta}(\boldsymbol{h}) = \frac{\dfrac{2\eta^2}{\pi}\left[\sqrt{1-\rho_1(\boldsymbol{h})^2} + \rho_1(\boldsymbol{h})\arcsin\rho_1(\boldsymbol{h}) - 1\right] + \sigma^2\rho(\boldsymbol{h})}{\sigma^2 + \eta^2(1 - 2/\pi)}. \tag{4}$$

For both models the location parameter is constant, $\mu(\boldsymbol{s}) = \mu$. The Gaussian and skew-Gaussian model parameters can be inspected with:

```
CorrParam("Matern")
NuisParam("Gaussian")
NuisParam("SkewGaussian")
```

A Matérn correlation model with `smooth=0.5` is exponential. Here `scale`=α, `nugget`=τ^2, `sill`=σ^2, and `skew`=η.

Estimation of anomalies data

For a Gaussian random vector $\boldsymbol{Y} = (Y(\boldsymbol{s}_1), \ldots, Y(\boldsymbol{s}_N))^\top$, with correlation matrix $\boldsymbol{R}$, the multivariate density is

$$f_{\boldsymbol{Y}}(\boldsymbol{y}; \boldsymbol{\theta}_Y) = (2\pi)^{-N/2}\,|\sigma^2\boldsymbol{R}|^{-1/2}\exp\left[-\frac{(\boldsymbol{y}-\mu\boldsymbol{1})^\top\boldsymbol{R}^{-1}(\boldsymbol{y}-\mu\boldsymbol{1})}{2\sigma^2}\right], \tag{5}$$

where $\boldsymbol{\theta}_Y = (\mu, \sigma^2, \alpha, \tau^2)^\top$; the smoothness is fixed. For 7,352 locations, maximizing the full multivariate likelihood can be demanding. We therefore use conditional pairwise composite likelihood (Bevilacqua and Gaetan, 2015). Similarly, for the skew-Gaussian vector $\boldsymbol{U}_\eta$, the bivariate density is of the form (Alegría et al., 2017)

$$f_{U_\eta;ij}(u_i, u_j; \boldsymbol{\theta}_{U_\eta}) = 2 \sum_{\ell=1}^{2} \phi_2(\boldsymbol{u}_{ij} - \boldsymbol{\mu}_{ij}; \boldsymbol{A}_\ell) \Phi_{n_\ell}(\boldsymbol{c}_\ell; \boldsymbol{0}, \boldsymbol{B}_\ell), \tag{6}$$

where the quantities $\boldsymbol{A}_\ell, \boldsymbol{B}_\ell, \boldsymbol{c}_\ell$ are defined in Alegría et al. (2017); $\boldsymbol{\theta}_{U_\eta} = (\mu, \sigma^2, \eta, \alpha, \tau^2)^\top$. The two estimators involve only bivariate and univariate marginal densities.

The conditional pairwise composite log-likelihoods are

$$\mathrm{pl}(\boldsymbol{\theta}_Y) = \sum_{i=1}^{N} \sum_{j \neq i} w_{ij} \big[\log f_{Y;ij}(y_i, y_j) - \log f_{Y;j}(y_j) \big] \tag{7}$$

and

$$\mathrm{pl}(\boldsymbol{\theta}_{U_\eta}) = \sum_{i=1}^{N} \sum_{j \neq i} w_{ij} \big[\log f_{U_\eta;ij}(u_i, u_j) - \log f_{U_\eta;j}(u_j) \big]. \tag{8}$$

Nearest-neighbour weights are generally directed, rather than symmetric:

$$w_{ij}(k) = \begin{cases} 1, & \boldsymbol{s}_i \in N_k(\boldsymbol{s}_j), \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

Here $N_k(\boldsymbol{s}_j)$ is the set of the k nearest neighbours of site $\boldsymbol{s}_j$. Both models are estimated through `GeoFit2`, using the `nlminb` optimizer and $k = 3$. The `GeoFit2` function first uses an independence composite likelihood step to initialize marginal parameters.

```
corrmodel="Matern"
I=Inf
start=list(mean=mean(z),sill=var(z),nugget=0.10,scale=200)
lower=list(mean=-I,sill=1e-6,nugget=0,scale=1e-6)
upper=list(mean=I,sill=I,nugget=1-1e-6,scale=I)
fixed=list(smooth=0.5)
pcl1=GeoFit2(coordx=loc,corrmodel=corrmodel,data=z,
 likelihood="Conditional",type="Pairwise",model="Gaussian",
 optimizer="nlminb",lower=lower,upper=upper,sensitivity=TRUE,
 neighb=3,start=start,fixed=fixed)
```

Bounds exclude degenerate sill/scale values and the forbidden endpoint `nugget=1`. This is a numerical safeguard, not a change of model.

The resulting historical Gaussian fit in the original 2025 tutorial was:

```
pcl1
 Maximum Composite-Likelihood Fitting of Gaussian Random Fields
 Setting: Conditional Composite-Likelihood
 Number of spatial coordinates: 7352
 Estimated parameters:
      mean    nugget      scale       sill
   0.03651   0.07004  112.69057    0.79248
```

We now fit the skew-Gaussian random field under the same Matérn covariance model.

```
start=list(nugget=0.1,scale=200,mean=mean(z),
           sill=var(z),skew=0.4)
lower2=list(mean=-I,sill=1e-6,skew=-I,scale=1e-6,nugget=0)
upper2=list(mean=I,sill=I,skew=I,scale=I,nugget=1-1e-6)
pcl2=GeoFit2(coordx=loc,corrmodel=corrmodel,data=z,
 likelihood="Conditional",type="Pairwise",model="SkewGaussian",
 sensitivity=TRUE,optimizer="nlminb",lower=lower2,
 upper=upper2,neighb=3,start=start,fixed=fixed)
pcl2
```

The skew-Gaussian field includes one additional asymmetry parameter.

The historical skew-Gaussian estimates printed in the original tutorial are:

```
 Maximum Composite-Likelihood Fitting of Skew Gaussian RF
 Setting: Conditional Composite-Likelihood
 Number of spatial coordinates: 7352
 Estimated parameters:
      mean      nugget      scale       sill      skew
  -0.63583    0.09506  156.00518    0.53084    0.84762
```

The estimated skewness parameter is *positive*, indicating right-skewness under the model representation in (3). The original tutorial incorrectly described it as negative. The maximized pairwise log-likelihood is higher for the more flexible skew-Gaussian field, which nests the Gaussian field at $\eta = 0$. Standard errors are computed by parametric bootstrap; this is computationally intensive for such a large data set.

```
set.seed(5)
pcl1_est=GeoVarestbootstrap(pcl1,K=100,
 optimizer="nlminb",lower=lower,upper=upper,parallel=TRUE)
set.seed(5)
pcl2_est=GeoVarestbootstrap(pcl2,K=100,
 optimizer="nlminb",lower=lower2,upper=upper2,parallel=TRUE)
```

These calls preserve 100 replications as in the original example.

The standard errors, confidence intervals and composite likelihood information criteria can be obtained from the fitted objects:

```
pcl1_est$stderr; pcl2_est$stderr
pcl1_est$conf.int; pcl2_est$conf.int
pcl1_est$claic; pcl2_est$claic
 CL-AIC values:
 Gaussian:       38473
 Skew-Gaussian:  38429
 GeoTests(pcl2_est,pcl1_est,statistic="Wald")
 Wald Chi-square: 58.11021 ; p= 2.478368e-14
```

In the historical example, CL-AIC favours the skew-Gaussian model. The Wald test for $H_0 : \eta = 0$ also rejects the Gaussian restriction.

Checking model assumptions

Given estimates of the model parameters, define standardized residuals for the Gaussian field as

$$\widehat{Y}^*(\boldsymbol{s}_i) = \frac{y(\boldsymbol{s}_i) - \widehat{\mu}}{\sqrt{\widehat{\sigma}^2}}, \qquad i = 1, \ldots, N. \tag{10}$$

For the skew-Gaussian field, note that μ is a *location* parameter rather than the marginal mean. Standardization must use the actual marginal mean and variance. Thus

$$\widehat{U}^*_\eta(\boldsymbol{s}_i) = \frac{u(\boldsymbol{s}_i) - \widehat{\mu} - \widehat{\eta}\sqrt{2/\pi}}{\sqrt{\widehat{\sigma}^2 + \widehat{\eta}^2(1-2/\pi)}}. \tag{11}$$

This is the convention used by `GeoResiduals` for real-valued fields: center by the estimated marginal mean and divide by the estimated marginal standard deviation. The resulting skew-Gaussian residuals have zero marginal mean and unit marginal variance, but not a Gaussian marginal distribution.

```
res1=GeoResiduals(pcl1)
res2=GeoResiduals(pcl2)
GeoQQ(res1,type="D",breaks=50,ylim=c(0,0.45))
GeoQQ(res2,type="D",breaks=50,ylim=c(0,0.45))
```

To check dependence, compare empirical and fitted semivariograms, calculated *separately* from the two residual objects:

```
vario1=GeoVariogram(data=res1$data,coordx=loc,maxdist=maxdist/4)
GeoCovariogram(res1,show.vario=TRUE,vario=vario1,pch=20,
               ylim=c(0,1.3))
vario2=GeoVariogram(data=res2$data,coordx=loc,maxdist=maxdist/4)
GeoCovariogram(res2,show.vario=TRUE,vario=vario2,pch=20,
               ylim=c(0,1.7))
```

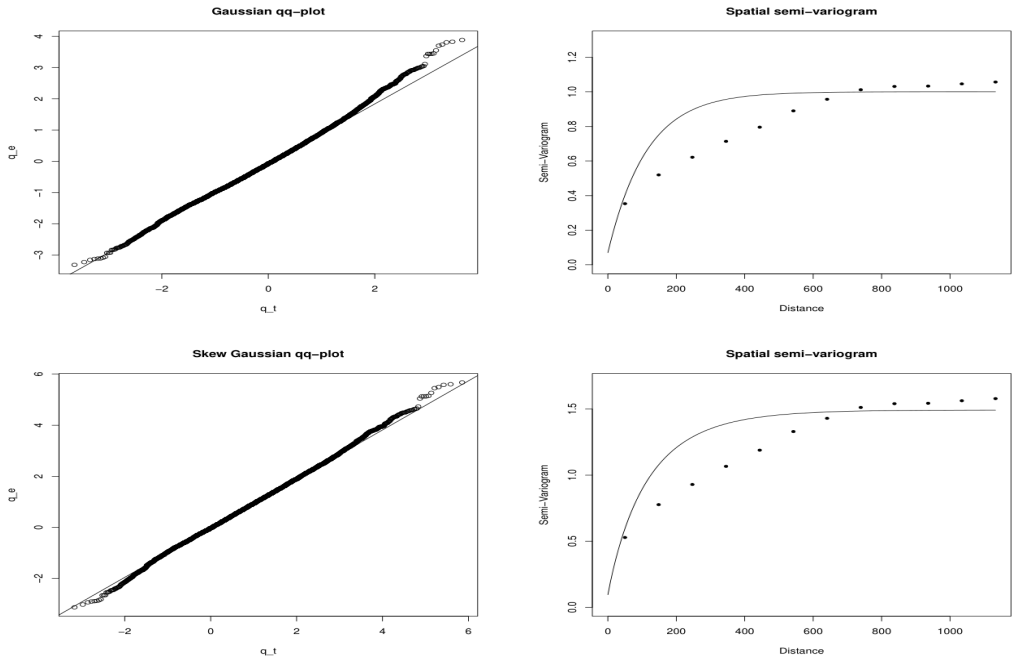


Figure 4: Historical Figure 4. Upper: Gaussian residual QQ plot and semivariogram. Lower: skew-Gaussian diagnostics.

The original graphs suggest closer agreement for the skew-Gaussian model. The graphs above are retained from the original tutorial; the corrected residual definitions should be used when regenerating them.

Prediction

The package `GeoModels` provides optimal linear prediction for Gaussian and skew-Gaussian random fields. For a spatial location $\boldsymbol{s}_0$, the best linear predictor of a field L is

$$\widehat{L}(\boldsymbol{s}_0) = m_L + \boldsymbol{c}_L^\top \boldsymbol{R}_L^{-1}(\boldsymbol{l} - m_L\mathbf{1}), \tag{12}$$

where $m_Y = \mu$ for the Gaussian model but $m_{U_\eta} = \mu + \eta\sqrt{2/\pi}$ for the skew-Gaussian model. Further, $\boldsymbol{c}_L = (\mathrm{cor}\{L(\boldsymbol{s}_0), L(\boldsymbol{s}_1)\}, \ldots, \mathrm{cor}\{L(\boldsymbol{s}_0), L(\boldsymbol{s}_N)\})^\top$ and $\boldsymbol{R}_L = [\mathrm{cor}\{L(\boldsymbol{s}_i), L(\boldsymbol{s}_j)\}]_{i,j=1}^N$. The mean squared prediction error is

$$\mathrm{MSE}\{\widehat{L}(\boldsymbol{s}_0)\} = \mathrm{Var}\{L(\boldsymbol{s}_0)\}(1 - \boldsymbol{c}_L^\top \boldsymbol{R}_L^{-1}\boldsymbol{c}_L). \tag{13}$$

The variance is σ^2 under the Gaussian model and $\sigma^2 + \eta^2(1 - 2/\pi)$ under the skew-Gaussian model. Prediction uses plug-in covariance and marginal moment estimates. In the Gaussian case, the linear predictor is also the conditional expectation; in general this equivalence does not hold for the skew-Gaussian field. The prediction performances are compared using cross-validation:

```
set.seed(9)
a1=GeoCV(pcl1,K=100,estimation=TRUE,n.fold=0.25,
         local=TRUE,neighb=100,parallel=TRUE)
set.seed(9)
a2=GeoCV(pcl2,K=100,estimation=TRUE,n.fold=0.25,
         local=TRUE,neighb=100,parallel=TRUE)
```

The cross-validation procedure uses 75% of the observations for model refitting and 25% for validation in each replication. Local prediction uses 100 nearest neighbours. The original tutorial reported:

```
mean(a1$rmse); mean(a2$rmse)
 Gaussian:      0.4808983
 Skew-Gaussian: 0.4804112
mean(a1$mae); mean(a2$mae)
 Gaussian:      0.3648200
 Skew-Gaussian: 0.3645137
```

The skew-Gaussian model performed only slightly better in predictive scores, even though its in-sample fit was considerably better.

For mapping, we first construct a grid of prediction sites in projected kilometres and then retain points within a valid geographic outline.

```
long1=min(loc[,1])-5;long2=max(loc[,1])+5
lat1=min(loc[,2])-5;lat2=max(loc[,2])+5
lat_seq=seq(lat1,lat2,23)
lon_seq=seq(long1,long2,23)
coords_tot=as.matrix(expand.grid(lon_seq,lat_seq))
gr.in <- locations.inside(coords_tot, SpP)
```

```
pr1=GeoKrigloc(pcl1,loc=gr.in,mse=TRUE,neighb=100)
pr2=GeoKrigloc(pcl2,loc=gr.in,mse=TRUE,neighb=100)
par(mfrow=c(2,2))
quilt.plot(gr.in,pr1$pred)
quilt.plot(gr.in,pr1$mse)
```

```
quilt.plot(gr.in,pr2$pred)
quilt.plot(gr.in,pr2$mse)
```

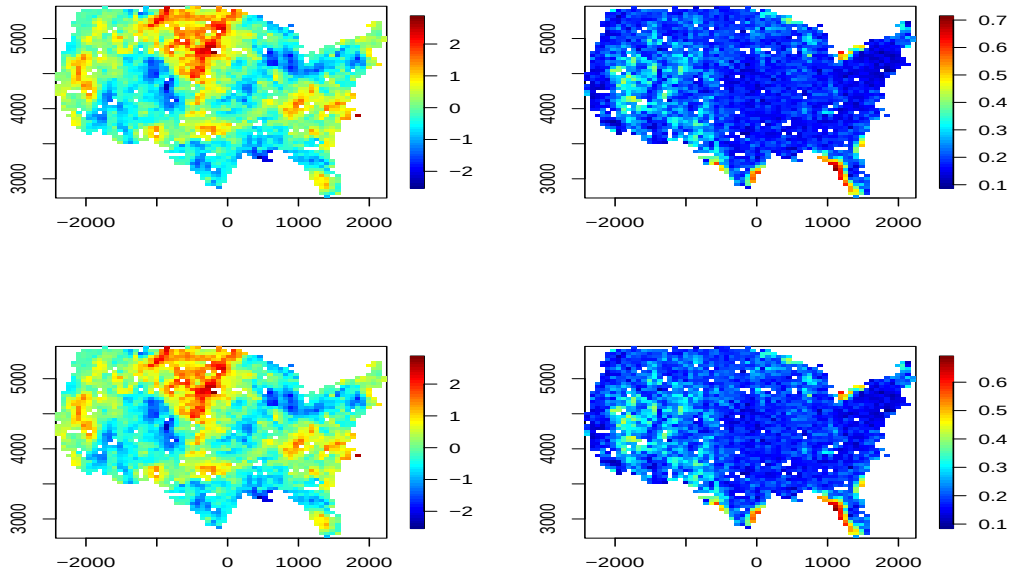


Figure 5: Gaussian (top row) and skew-Gaussian (bottom row) kriging predictions and associated MSE maps.

References

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Hao Zhang and Abdel H. El-Shaarawi. On spatial skew-Gaussian processes and applications. *Environmetrics*, 21(1):33–47, 2010.